Enterprise Banking Stress Testing Platform

V5 Statistical Risk Modelling

Module 01 - Enterprise Portfolio Validation

Dataset Summary

- Retail Credit Portfolio: 100,000 Records
- Wholesale Credit Portfolio: 100,000 Records
- Total Portfolio: 200,000 Records

Purpose

- Read SAS-generated Enterprise Portfolio Dataset
- Perform Data Validation
- Verify Portfolio Distribution
- Perform Descriptive Analysis
- Support SAS to Python Reconciliation

In [13]:
```
# ============================================================
# IMPORT LIBRARIES
# ============================================================

import pandas as pd
import numpy as np
import time
from datetime import datetime

# ============================================================
# EXECUTION START
# ============================================================

start_time = time.time()

print("=" * 60)
print("V5 - ENTERPRISE PORTFOLIO VALIDATION")
print("=" * 60)
print(f"Execution Started : {datetime.now()}")
print("=" * 60)
```

```
============================================================
V5 - ENTERPRISE PORTFOLIO VALIDATION
============================================================
Execution Started : 2026-10-30 06:08:39.240180
============================================================
```

In [14]:
```
# ============================================================
# READ SAS GENERATED DATASET
```

```
# ============================================================

file_path = "../DATA/enterprise_portfolio_200k.csv"

df = pd.read_csv(file_path)

print("Dataset Loaded Successfully")
print(f"Rows    : {df.shape[0]:,}")
print(f"Columns : {df.shape[1]:,}")
```

```
Dataset Loaded Successfully
Rows    : 200,000
Columns : 24
```

In [15]:
```
# ============================================================
# COLUMN VALIDATION
# ============================================================

print("=" * 60)
print("COLUMN VALIDATION")
print("=" * 60)

for col in df.columns:
    print(col)
```

```
============================================================
COLUMN VALIDATION
============================================================
CUSTOMER_ID
LOAN_ID
PORTFOLIO_TYPE
PRODUCT_TYPE
REGION
DELINQUENCY_BUCKET
RECORD_ID
PRODUCT_RANDOM
REGION_RANDOM
CREDIT_SCORE
ANNUAL_INCOME
LOAN_AMOUNT
OUTSTANDING_BALANCE
LTV
DTI
YEARS_ON_BOOK
RANDOM_DPD
GDP_GROWTH_RATE
UNEMPLOYMENT_RATE
INTEREST_RATE
PD
LGD
EAD
OBSERVED_DEFAULT
```

In [16]:
```
# ============================================================
# PORTFOLIO DISTRIBUTION
# ============================================================
```

```
portfolio_dist = (
    df["PORTFOLIO_TYPE"]
    .value_counts()
)

print(portfolio_dist)
```

```
PORTFOLIO_TYPE
RETAIL       100000
WHOLESALE    100000
Name: count, dtype: int64
```

In [17]:
```
# ============================================================
# PRODUCT DISTRIBUTION
# ============================================================

product_dist = (
    df["PRODUCT_TYPE"]
    .value_counts()
    .sort_index()
)

print(product_dist)
```

```
PRODUCT_TYPE
Auto Loan          25133
Corporate Loan     24989
Credit Card        24944
Mortgage           24903
Personal Loan      25020
Project Finance    24983
SME Loan           24896
Trade Finance      25132
Name: count, dtype: int64
```

In [18]:
```
# ============================================================
# MISSING VALUE VALIDATION
# ============================================================

missing_values = df.isnull().sum()

print(
    missing_values[
        missing_values > 0
    ]
)
```

```
Series([], dtype: int64)
```

In [19]:
```
# ============================================================
# DESCRIPTIVE STATISTICS
# ============================================================

numeric_cols = [
    "CREDIT_SCORE",
    "ANNUAL_INCOME",
    "LOAN_AMOUNT",
```

```
    "OUTSTANDING_BALANCE",
    "LTV",
    "DTI",
    "PD",
    "LGD",
    "EAD"
]

df[numeric_cols].describe()
```

Out[19]:

	CREDIT_SCORE	ANNUAL_INCOME	LOAN_AMOUNT	OUTSTANDING_BALANCE	
count	200000.000000	2.000000e+05	2.000000e+05	2.000000e+05	20000
mean	574.375315	2.801571e+06	5.035270e+06	3.027323e+06	6
std	159.036618	1.444199e+06	2.883985e+06	2.192060e+06	1
min	300.000000	3.000000e+05	5.000000e+04	1.080000e+04	4
25%	436.000000	1.551000e+06	2.536300e+06	1.264600e+06	5
50%	575.000000	2.802300e+06	5.036000e+06	2.548950e+06	7
75%	713.000000	4.051000e+06	7.525000e+06	4.412700e+06	8
max	849.000000	5.300000e+06	1.005000e+07	1.003210e+07	10

In [20]:

```
# ============================================================
# SAMPLE PORTFOLIO RECORDS
# ============================================================

df.head(10)
```

Out[20]:

	CUSTOMER_ID	LOAN_ID	PORTFOLIO_TYPE	PRODUCT_TYPE	REGION	DELINQUEN
0	CUST00000001	LOAN00000001	RETAIL	Auto Loan	Central	9(
1	CUST00000002	LOAN00000002	RETAIL	Mortgage	West	
2	CUST00000003	LOAN00000003	RETAIL	Credit Card	East	
3	CUST00000004	LOAN00000004	RETAIL	Credit Card	South	
4	CUST00000005	LOAN00000005	RETAIL	Auto Loan	North	
5	CUST00000006	LOAN00000006	RETAIL	Auto Loan	Central	
6	CUST00000007	LOAN00000007	RETAIL	Mortgage	East	
7	CUST00000008	LOAN00000008	RETAIL	Personal Loan	South	
8	CUST00000009	LOAN00000009	RETAIL	Mortgage	South	9(
9	CUST00000010	LOAN00000010	RETAIL	Personal Loan	East	

10 rows × 24 columns

In [21]:
```
# ============================================================
# EXECUTION SUMMARY
# ============================================================

end_time = time.time()

duration = round(
    end_time - start_time,
    2
)

print("=" * 60)
print("VALIDATION COMPLETED")
print("=" * 60)

print(f"Total Records      : {len(df):,}")
print(f"Execution Duration : {duration} seconds")

print("=" * 60)
```

```
============================================================
VALIDATION COMPLETED
============================================================
Total Records      : 200,000
Execution Duration : 207.08 seconds
============================================================
```

In [22]:
```
validation_summary = """
============================================================
V5 ENTERPRISE PORTFOLIO VALIDATION COMPLETED
============================================================
```

```
Dataset Size : 200,000

Retail Portfolio : 100,000

Wholesale Portfolio : 100,000

Status : SUCCESS

============================================================
"""
print(validation_summary)
```

```
============================================================
V5 ENTERPRISE PORTFOLIO VALIDATION COMPLETED
============================================================

Dataset Size : 200,000

Retail Portfolio : 100,000

Wholesale Portfolio : 100,000

Status : SUCCESS

============================================================
```